



Motion Sequencing Reveals Persistent Sex-Specific Behavioral Outcomes Following Chronic Corticosterone Exposure

Poster #D8

Kaitlyn Huynh¹, Mohan Yang², Carlos Rodriguez-Ortiz¹, Steven Grieco¹, Krisha Thakker¹, Zhaoxia Yu², Xiangmin Xu¹

¹Department of Anatomy & Neurobiology, School of Medicine, University of California, Irvine, Irvine, CA, USA

²Department of Statistics, Donald Bren School of Information and Computer Sciences, University of California, Irvine, Irvine, CA, USA



Abstract

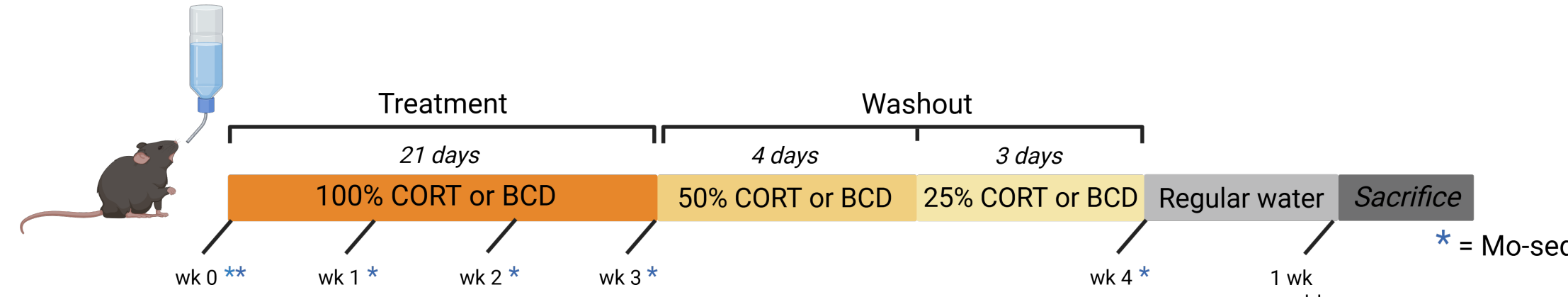
Chronic psychological stress produces persistent behavioral adaptations, yet the temporal dynamics of these changes are not well defined, particularly across sexes. Traditional behavioral assays often quantify discrete behavioral endpoints and may overlook subtle changes in the organization of spontaneous behavior.¹ Motion Sequencing (MoSeq) resolves spontaneous behavior into sub-second behavioral syllables, enabling longitudinal analysis of behavioral structure throughout chronic corticosterone (CORT) exposure and recovery. We used MoSeq to examine changes across behavioral domains including exploration, locomotion, self-directed behaviors, and immobility in male and female mice. Because spontaneous behavior is organized into structured sequences rather than isolated actions, syllable usage provides a sensitive measure of how chronic stress reorganizes behavioral dynamics. Given known sex differences in stress susceptibility and adaptation², we performed sex-specific analyses to identify transient and persistent behavioral adaptations across baseline, treatment, and washout periods. These findings demonstrate the utility of unsupervised behavioral phenotyping for detecting stress-induced behavioral changes that may be overlooked by conventional behavioral assays.

Experimental Design

A Cohort composition

Sex	BCD	CORT
♂ Male	n = 23	n = 20
♀ Female	n = 12	n = 12

B Treatment paradigm



C MoSeq Workflow

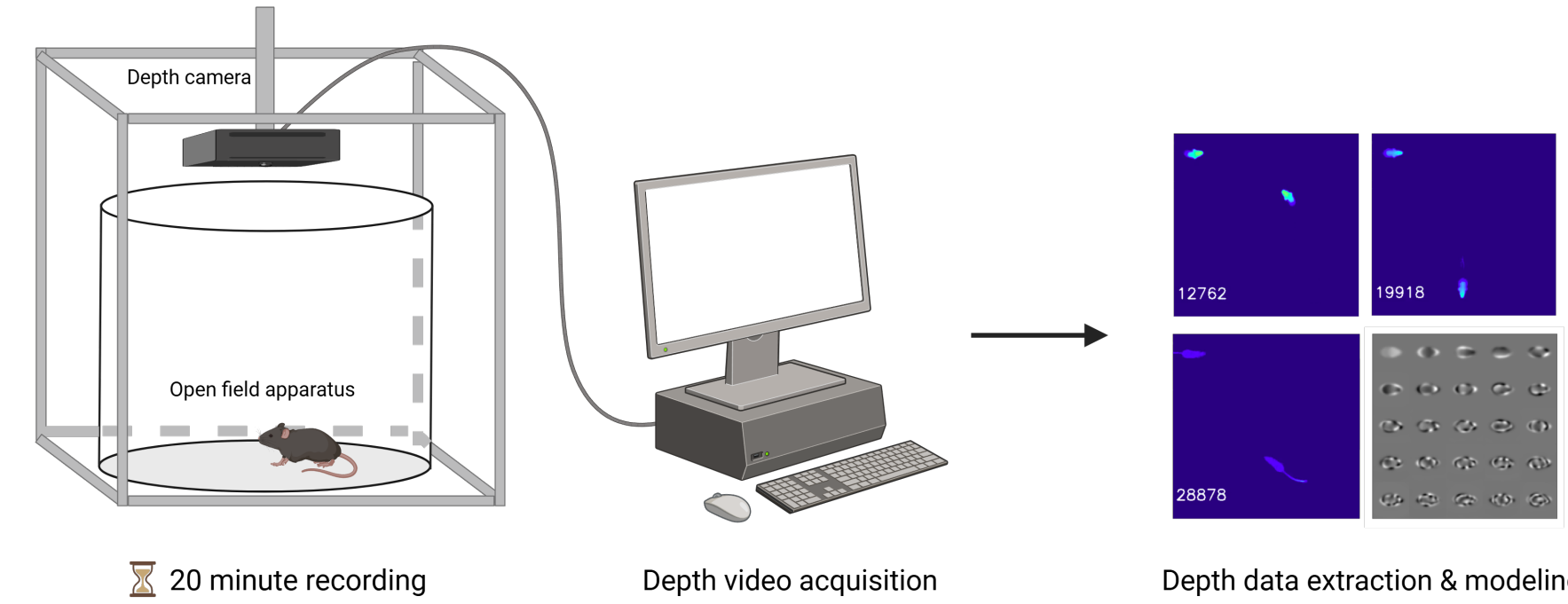


Figure 1. Experimental design for treatment cohorts & MoSeq workflow.

(A) Cohort composition by sex and treatment group. (B) Experimental timeline illustrating chronic corticosterone (CORT) treatment, washout, and MoSeq recording timepoints. (C) MoSeq acquisition workflow, from depth-video recording to behavioral modeling.

Why MoSeq?

- Unsupervised behavioral analysis
- Sub-second behavioral dynamics
- Longitudinal tracking of spontaneous behavior
- Detects subtle behavioral organization

Results

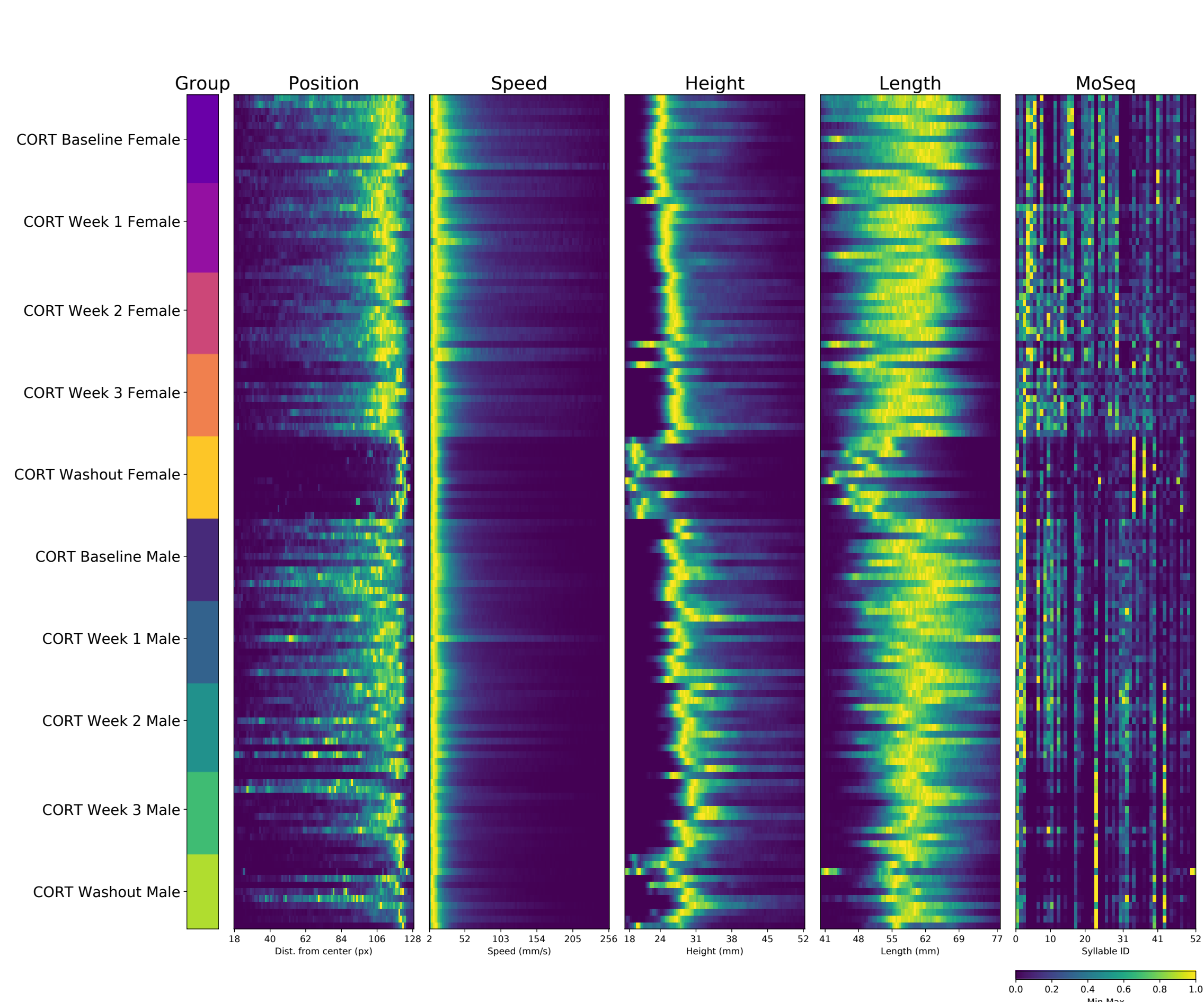


Figure 2. MoSeq kinematic fingerprint across sex & CORT treatment timepoints. Behavioral fingerprints summarize kinematic features extracted from MoSeq analysis, including distance from center, speed, length, height, and syllable usage probability, visualized across sex and experimental timepoints.

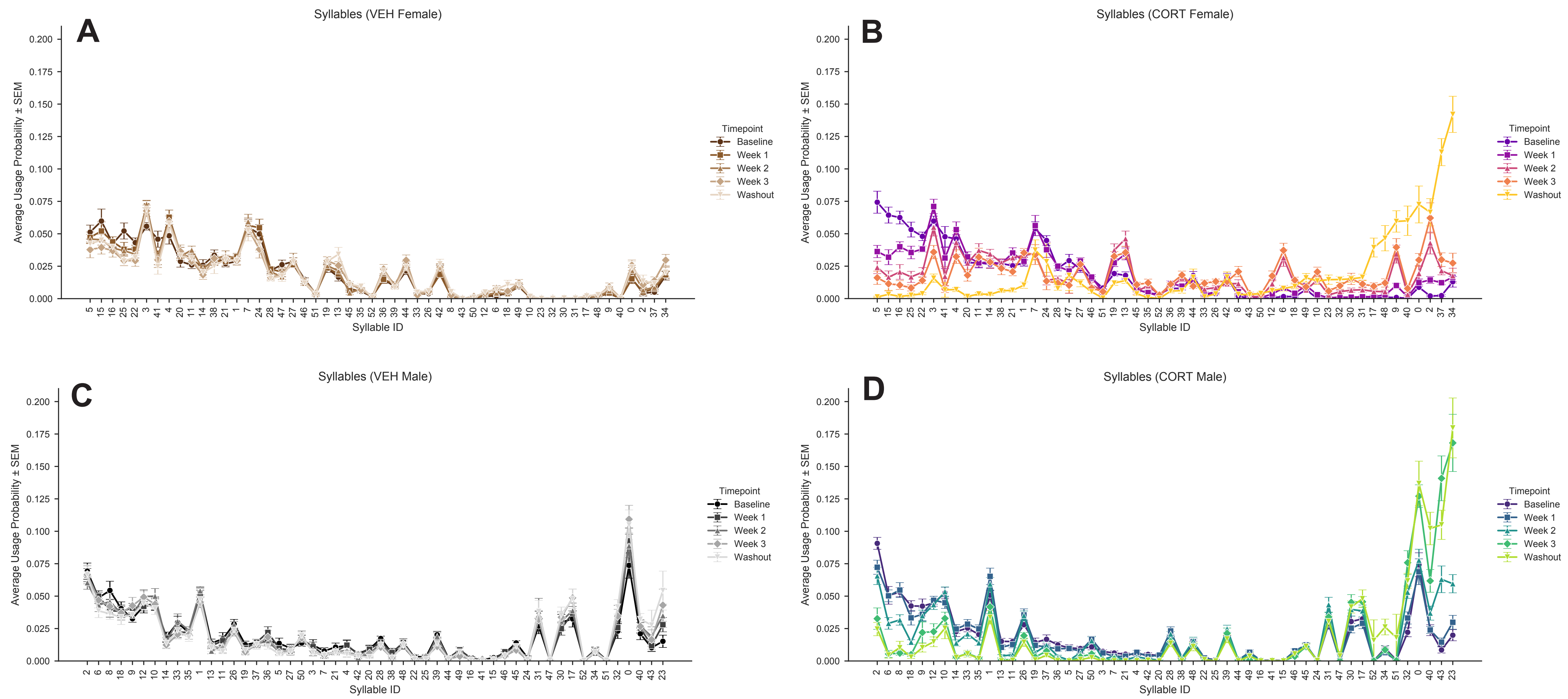


Figure 4. Average syllable usage probability across sex & CORT treatment timepoints. Average behavioral syllable usage probability across baseline, CORT treatment timepoints, and washout. Panels show female vehicle (A), female CORT (B), male vehicle (C), and male CORT (D).

Behavioral domains

Representative MoSeq depth frame sequence

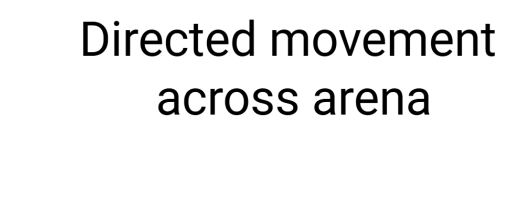
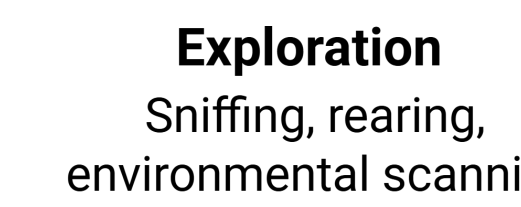
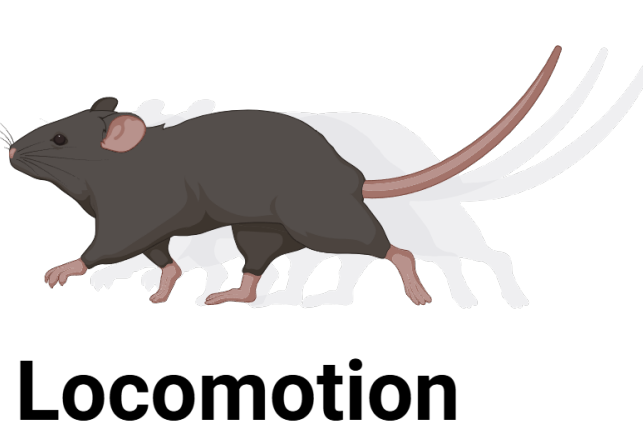


Figure 3. MoSeq behavioral syllable classification into ethological domains. Representative frames extracted from MoSeq-generated syllable videos illustrate five manually annotated behavioral domains. MoSeq identified recurring movement patterns through unsupervised analysis that were subsequently grouped based on observed behaviors. Kinect depth visualizations depict body position relative to the sensor; warmer colors indicate regions proximal to the Kinect, and cooler colors distal.



Locomotion



Immobility

Figure 5. Average usage probability of locomotion-associated syllables. Usage is plotted across baseline, CORT treatment timepoints, and washout. Panels show female VEH (A), female CORT (B), male VEH (C), and male CORT (D).

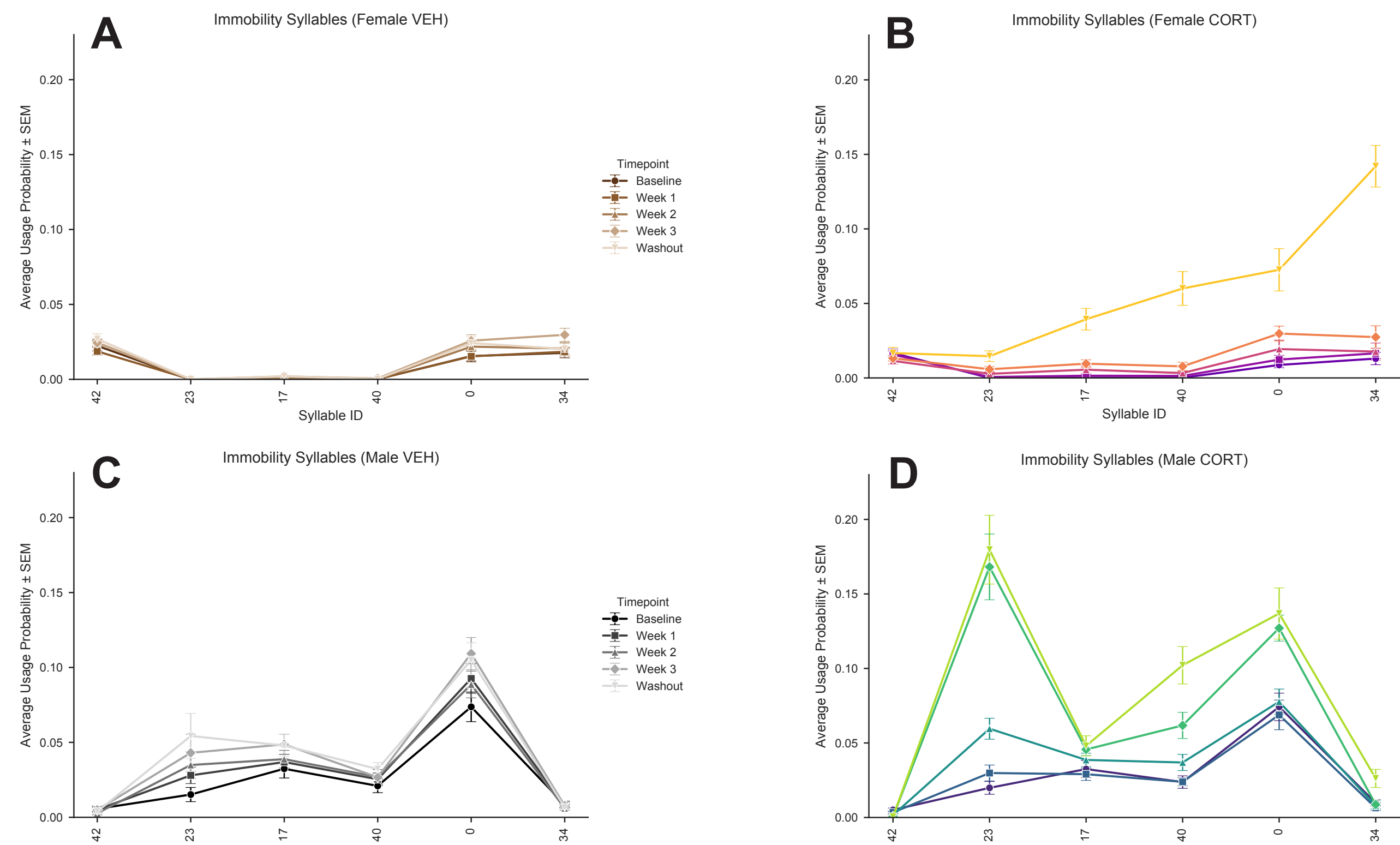
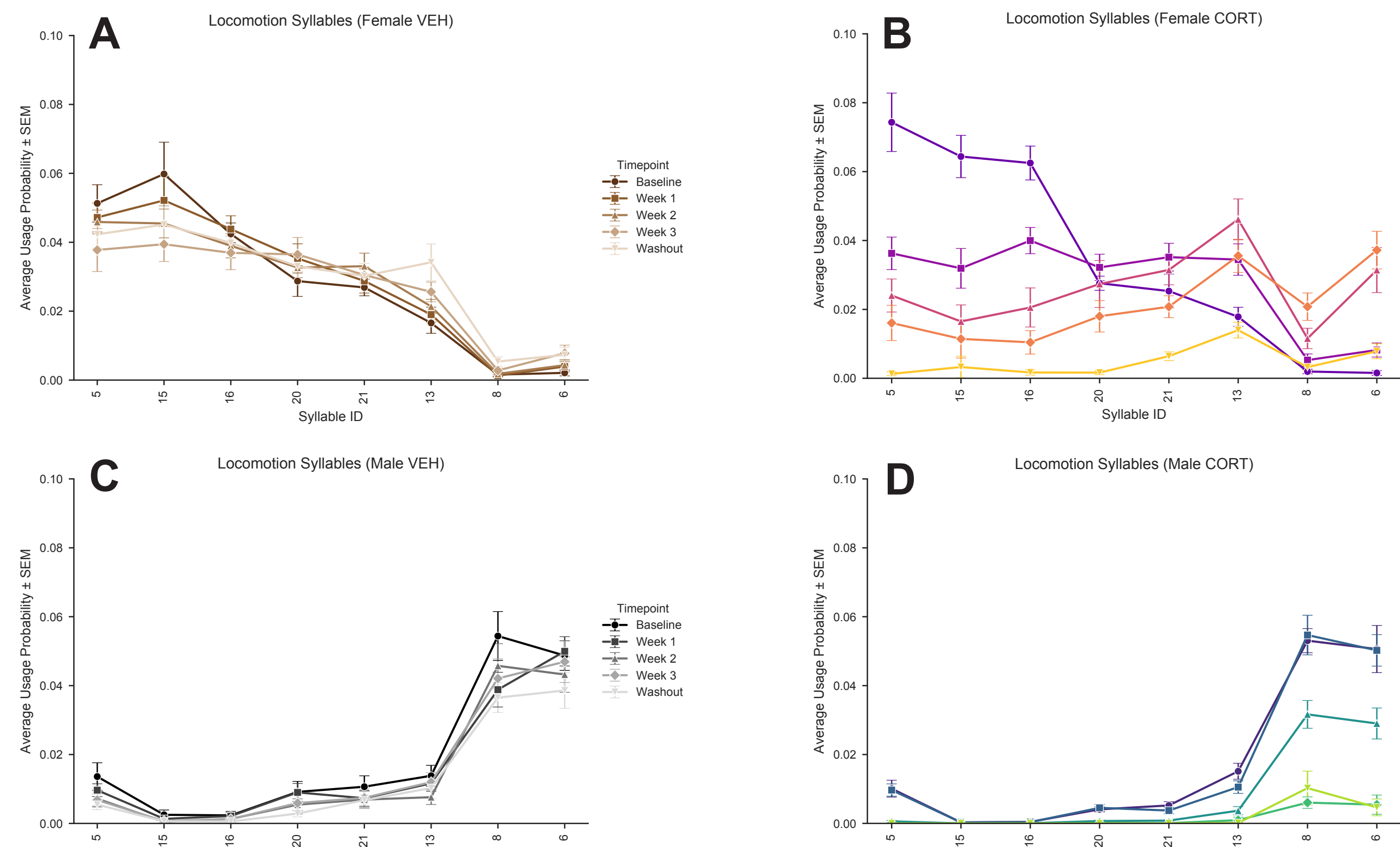


Figure 6. Average usage probability of immobility-associated syllables. Usage is plotted across baseline, CORT treatment timepoints, and washout. Panels show female VEH (A), female CORT (B), male VEH (C), and male CORT (D).

Conclusions

- Chronic corticosterone exposure alters spontaneous behavioral organization, producing distinct behavioral profiles captured through MoSeq analysis.
- These behavioral adaptations differ between male and female mice, demonstrating sex-specific responses to prolonged glucocorticoid exposure.
- MoSeq enables identification of subtle changes in individual behavioral syllables and domains that may not be detected through conventional behavioral assays.
- Together, these findings highlight the importance of high-resolution behavioral profiling and consideration of sex differences when investigating the effects of chronic stress.

Acknowledgements

References

- ¹ Lin et al., 2024 — MoSeq methodology and analysis framework
- ² Wiltshcko et al., 2015 — Original MoSeq approach and behavioral syllable discovery